

Problem 33. Let $a > b > 0$. Prove that

$$\sqrt{2}a^3 + \frac{3}{ab-b^2} \geq 10.$$

When does equality hold?

Solution. Observe that, by the AM-GM Inequality,

$$ab - b^2 = (a - b)b \leq \left(\frac{a - b + b}{2}\right)^2 = \frac{a^2}{4}$$

and so, by the AM-GM Inequality (again), we have

$$\sqrt{2}a^3 + \frac{3}{ab-b^2} \geq \sqrt{2}a^3 + \frac{12}{a^2} = \frac{a^3}{\sqrt{2}} + \frac{a^3}{\sqrt{2}} + \frac{4}{a^2} + \frac{4}{a^2} + \frac{4}{a^2} \geq 5\left(\frac{a^6 \cdot 4^3}{2a^6}\right)^{\frac{1}{5}} = 10.$$

Equality holds if and only if $a - b = b$ and $\frac{a^3}{\sqrt{2}} = \frac{4}{a^2}$, that is, $a = \sqrt{2}$, $b = \frac{\sqrt{2}}{2}$. □

Problem 34. Let a, b, c be non-zero real numbers such that

$$\begin{cases} a^2 + b + c = \frac{1}{a}, \\ b^2 + c + a = \frac{1}{b}, \\ c^2 + a + b = \frac{1}{c}. \end{cases}$$

Prove that at least two of a, b, c are equal.

Solution 1. Assume the contrary, that is, a, b, c are pairwise distinct. We subtract the second equation from the first equation, move everything to the left-side to obtain

$$(a - b)\left(a + b - 1 + \frac{1}{ab}\right) = a^2 + b + c - (b^2 + c + a) - \left(\frac{1}{a} - \frac{1}{b}\right) = 0.$$

Since $a \neq b$, we have $a + b - 1 + \frac{1}{ab} = 0$. Similarly, since $b \neq c$, we have $b + c - 1 + \frac{1}{bc} = 0$. Subtract this from the previous one and get

$$(a - c)\left(1 - \frac{1}{abc}\right) = a + b - 1 + \frac{1}{ab} - \left(b + c - 1 + \frac{1}{bc}\right) = 0.$$

Since $a \neq c$, we have $abc = 1$. Now observe that

$$a + b - 1 + c = a + b - 1 + \frac{abc}{ab} = 0$$

and so $b + c = 1 - a$. Put this into the original first equation to obtain

$$(a - 1)\left(\frac{a^2 + 1}{a}\right) = (a - 1)\left(a + \frac{1}{a}\right) = a^2 + 1 - a - \frac{1}{a} = a^2 + b + c - \frac{1}{a} = 0.$$

Since $a \neq 0$ and $a^2 + 1 \neq 0$, we have $a = 1$. Similarly, we have $b = c = 1$. This contradicts the assumption. Therefore, at least two of a, b, c are equal. □

Solution 2. Again, we assume the contrary, that is, a, b, c are pairwise distinct. We rewrite the given equations as

$$a^3 + ab + ac = b^3 + bc + ba = c^3 + ca + cb = 1.$$

Now observe that

$$(a - b)(a^2 + ab + b^2 + c) = a^3 + ab + ac - (b^3 + bc + ba) = 0.$$

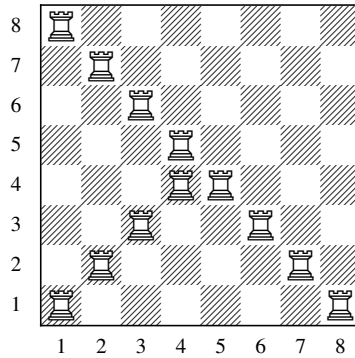
Since $a \neq b$, we have $a^2 + ab + b^2 + c = 0$. Similarly, we have $b^2 + bc + c^2 + a = 0$. We subtract this from the previous one and get

$$(a - c)(a + c + b - 1) = a^2 + ab + b^2 + c - (b^2 + bc + c^2 + a) = 0,$$

and since $a \neq c$, we have $a + c + b = 1$. Then we proceed as in **Solution 1**. □

Problem 35. Each square of $n \times n$ chessboard either contains a rook or is empty. Suppose that for any two rooks not threatening each other there is an empty square which is threatened by each of these rooks. Find the maximal possible number of rooks on the chessboard.

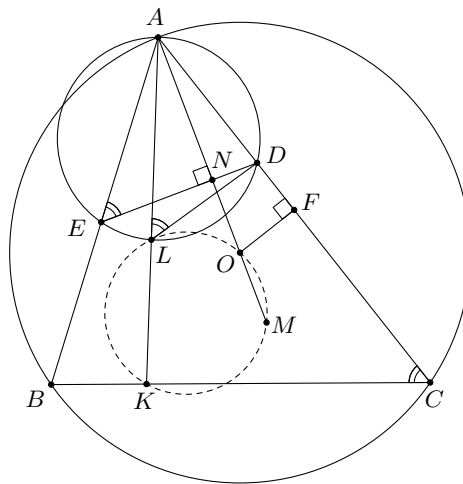
Solution. Each column contains at most two rooks, otherwise the uppermost and lowermost rooks on this column do not satisfy the conditions. We consider two columns each containing two rooks. Let a_1, a_2 and b_1, b_2 be rooks on the first and second columns, respectively (a_1 is above a_2 and b_1 is above b_2). We will show that any two of these rooks cannot be on the same row. We assume the contrary, that is, a_1 and b_1 are on the same row. Without loss of generality we suppose that the row containing a_2 is not above the row containing b_2 . Then a_2 and b_1 do not satisfy the conditions. Now suppose that a_1 and b_2 are on the same row. Then a_2 and b_1 do not satisfy the conditions. The cases when a_2 and b_2 are on the same row or a_2 and b_1 are on the same row are similar. This completes the proof. (Similar arguments show that there are only two possibilities: either both rows containing b_1 and b_2 are between the rows containing a_1 and a_2 , or vice-versa, but this fact is not used in the proof.) Now by the Pigeonhole Principle the total number of columns containing two rooks is at most $\lfloor n/2 \rfloor$. Therefore, even if each remaining column contains one rook, the total number of rooks on the board will be at most $2 \lfloor n/2 \rfloor + n - \lfloor n/2 \rfloor = \lfloor 3n/2 \rfloor$.



Now we give an example for $\lfloor 3n/2 \rfloor$ rooks satisfying the conditions. We enumerate the rows by $1, 2, \dots, n$ starting from the lowest row and enumerate the columns by $1, 2, \dots, n$ starting from the leftmost column. (the configuration above shows the case $n = 8$.) Let the square on the intersection of column i and row j be (i, j) . For each $1 \leq i \leq n$, we place a rook on the square $(i, n + 1 - i)$ (n rooks in total) and for each $1 \leq i \leq \lfloor n/2 \rfloor$ we place a rook to the square (i, i) ($\lfloor n/2 \rfloor$ rooks in total). This configuration satisfies the conditions. $\square$

Problem 36. Let O be the circumcentre of an acute triangle ABC . A line perpendicular to AO intersects the line segments AC and AB at D and E , respectively. Let K be a point on BC which is not on the line AO . The line AK intersects the circumcircle of $\triangle ADE$ again at L . Let M be the point symmetric to A with respect to the line DE . Show that the points K, L, M, O are concyclic.

Solution.



Let AM and DE intersect at N , and let F be the foot of the perpendicular line from O to AC . Since $AO \perp DE$ and $\angle BAO = 90^\circ - \angle ACB$, we have

$$\angle ALD = \angle AED = 90^\circ - \angle EAN = 90^\circ - \angle BAO = \angle ACB$$

and hence the points K, C, D, L are concyclic and so

$$AL \cdot AK = AD \cdot AC.$$

Now, since $\angle DNO = \angle OFD = 90^\circ$, then D, N, O, F are concyclic and so

$$AD \cdot AF = AN \cdot AO.$$

Therefore, since F and N are, respectively, the midpoints of AC and AM , we have

$$AL \cdot AK = AD \cdot AC = AD \cdot (2AF) = (2AN) \cdot AO = AM \cdot AO,$$

and so K, L, M, O are concyclic.

□